

Subject: *Data Quality Findings & Key Insights from Fetch Analysis*

Hello Everyone,

I hope this message finds you well. I've conducted an in-depth analysis of our transaction, product, and user data, identifying key data quality issues, an interesting trend, and a few outstanding questions that require further investigation. Below is a summary of my findings and next steps.

Key Data Quality Issues

1. Missing Data:

- CATEGORY_4 in the **Products** table is missing for **92% of records**, making it difficult to assess its usefulness.
- LANGUAGE is missing for **30% of users**, potentially impacting user segmentation.
- BARCODE is missing for **~11.5% of transactions**, which could lead to incomplete sales insights.

2. Data Inconsistencies:

- FINAL_QUANTITY contains values like "zero" instead of 0, leading to inconsistencies in purchase volume calculations.
- BARCODE is stored as a **float**, potentially dropping leading zeros and causing mismatches across datasets.
- **Duplicate records** were found in the **Products (215) and Transactions (171) tables**, which may skew aggregate analyses.

3. Unusual Values:

- **Negative barcode values** exist in some transactions, which are likely erroneous.
- **9,931 unexpected gender values**, suggesting inconsistencies in user demographics.

Interesting Trend Identified

One standout insight is the **dominance of Walmart** in transaction volume.

- Walmart accounts for the **largest share of scanned receipts**, significantly ahead of other retailers like Dollar General and ALDI.
- Understanding why Walmart is a preferred retailer for Fetch users could provide opportunities for **targeted promotions and partnerships**.

Outstanding Questions & Request for Action

To further improve data integrity and extract actionable insights, I'd appreciate assistance with the following:

1. Missing Data Resolution

- Can we confirm whether missing **CATEGORY_4** values indicate an unknown classification or if data was lost during ingestion?
- Should we assume missing LANGUAGE values default to "**en**", or is there a data collection gap?

2. Data Standardization & Fixes

- Can we correct FINAL_QUANTITY formatting to ensure numerical consistency?
- Should we treat BARCODE as a **string** instead of a float to prevent data loss?

3. Strategic Business Input

- Given Walmart's dominance, should we explore **retail-specific incentives** to further engage Fetch users at key partner locations?

I'd love to discuss these findings in more detail and align on next steps. Let me know when we can connect!

Best Regards,
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